Fiscal Year:	FY 2014	**Task Last Updated:**	FY 10/23/2014
PI Name:	Delp, Michael Ph.D.		
Project Title:	Regional Arterial Remodeling Induced by Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology (2) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	mdelp@fsu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	352-214-1195
Organization Name:	Florida State University		
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City:	Tallahassee	**State:**	FL
Zip Code:	32306-0001	**Congressional District:**	2
Comments:	Previous affiliations were University of Florida (mid-2007-June 2014), West Virginia University (mid-2005 to mid-2007), and Texas A&M University (1995 to mid-2005).		
Project Type:	Flight	**Solicitation / Funding Source:**	2007 Space Flight: Bion-M1 NNH07ZTT001N
Start Date:	08/01/2009	**End Date:**	07/31/2014
No. of Post Docs:	0	**No. of PhD Degrees:**	3
No. of PhD Candidates:	5	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
Contact Email:	jeffrey.d.smith2@nasa.gov		
Flight Program:	Bion/Shuttle		
Flight Assignment:	STS-131 , STS-133 , STS-135 , Bion M1 NOTE: Extended to 7/31/2014 per A. Chu/ARC (Ed., 4/30/2014)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):			
Grant/Contract No.:	NNX09AP06G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Tail suspension or hindlimb unloading of rodents has been used as a ground-based model of microgravity because tail suspension, like actual microgravity, 1) induces a headward shift of fluids and 2) diminishes the contractile activity of hindlimb skeletal muscles that are normally involved in maintaining posture. It is now evident that the shift of fluid toward the head and the unloading of postural muscles alter the mechanical forces exerted on arteries, the vessels responsible for regulating tissue blood flow and arterial blood pressure. For example, the smooth muscle cells in the small arteries of hindlimb skeletal muscle atrophies with tail suspension, and results in a diminished ability of these vessels to vasoconstrict. The compromised ability to constrict reduces the ability of the rats to maintain normal blood pressure, and consequently, these animals are susceptible to abnormal reductions in arterial pressure when the body position of the animal is moved from a horizontal position to a 70o head-up position. Conversely, cerebral arteries in the head hypertrophy, which can limit brain blood flow and result in loss of consciousness during head-up tilt. The purpose of this proposal is to determine whether the fluid shifts and reduced activity of postural muscles that are induced by microgravity similarly alter rodent arterial vessel structure and some key cellular signaling pathways. In other words, does space flight result in smooth muscle atrophy of peripheral arteries and hypertrophy of cerebral arteries? This work will provide potentially important information about the mechanisms underlying the reduced fitness and the inability to properly regulate blood pressure of astronauts returning to Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These experiments will contribute toward attaining a better understanding of how fundamental biological systems, such as the cardiovascular system, respond to the microgravity environment. The detailed study of the resulting vascular adaptations triggered by microgravity will yield essential information on the basic physiological responses of individual blood vessels involved in blood flow and pressure regulation. Further, analysis of cerebral arteries will provide potentially important information about the visual impairment of astronauts induced by microgravity This information will also support the development of treatments or countermeasures to improve crew health and performance following their return to a gravitational environment.
Task Progress:	In the cerebral (basilar) arteries from the Bion-M1flight, vasoconstriction elicited through a voltage- gated Ca2+ mechanism (KCl) and thromboxane A2 receptors (U46619) were lower in crebral artieries from spaceflight mice. Inhibition of Rho-kinase activity (Y27632) abolished group differences in U46619-evoked contractions. Endothelium-dependent vasodilation elicited by acetylcholine was lower in cerebral arteries from flight mice. The pressure-diameter rlation was lower in arteries from flight mice relative to that in habitat control mice. This decrease in vascular distensibility did not appear to be related to differences in the extracellular matrix protein elastin or collagen content or the lastin/collagen ratio in the basilar arteries. Diameter, medial wall thickness, and medial cross-sectional area of unpressurized basilar arteries were not different among groups. A publication on the Bion results has been submitted: Sofronova, SI, OS Tarasova, D Gaynullina, AA Borzykh, BJ Behnke, JN Stabley, DJ McCullough, JJ Maraj, M Hanna, B Chen, JM Muller-Delp, OL Vinogradova, and MD Delp. Spaceflight on the Bion-M1 biosatellite alters cerebral artery vasomotor and mechanical properties in mice. Submitted. [Ed. note 2015--article now published ; see Bibliography citation]
Bibliography Type:	Description: (Last Updated: 07/09/2025)
Articles in Peer-reviewed Journals	Sofronova SI, Tarasova OS, Gaynullina D, Borzykh AA, Behnke BJ, Stabley JN, McCullough DJ, Maraj JJ, Hanna M, Chen B, Muller-Delp JN, Vinogradova OL, Delp MD. "Spaceflight on the Bion-M1 biosatellite alters cerebral artery vasomotor and mechanical properties in mice." Journal of Applied Physiology. 2015 Apr 1;118(7):830-8. Epub 2015 Jan 15. http://dx.doi.org/10.1152/japplphysiol.00976.2014 ; PMID: 25593287; PMCID: PMC4385880 [Erratum in: J Appl Physiol (1985). 2016 May 15;120(10):1267.] [Note also--reported in October 2014 as Submitted] , Apr-2015
Articles in Peer-reviewed Journals	Tarasova OS, Kalenchuk VU, Borovik AS, Golubinskaya VO, Delp MD, Vinogradova OL. "Simulated microgravity induces regionally distinct neurovascular and structural remodeling of skeletal muscle and cutaneous arteries in the rat." Front Physiol. 2020 Jun 30;11:675. https://www.frontiersin.org/article/10.3389/fphys.2020.00675 ; PMID: 32695017; PMCID: PMC7339929 , Jun-2020